BTS-IV(S)-04.22-0234 Reg. No. | | | | | | | | |

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B.Tech. Degree IV Semester Supplementary Examination April 2022

ME 15-1403 MECHATRONICS
(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

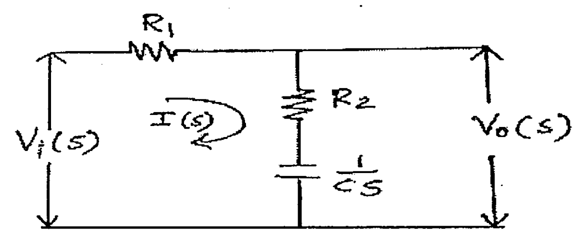
PART A
(Answer *ALL* questions)

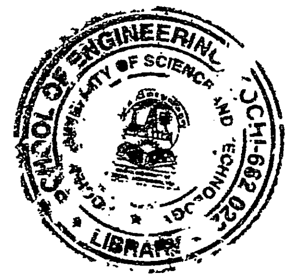
$(10 \times 2 = 20)$

I. (a) Explain the working of a gear pump.

(b) Describe strain gauge.

(c) Obtain the transfer function of the electrical network shown in figure.





(d) A unity feedback system is characterized by an open loop transfer function $G(s) = \dfrac{K}{S(S+10)}$. Determine the gain *K*, so that system will have a damping ratio of 0.5 for this value of *K*.

(e) Describe PI controller.

(f) Describe the building blocks for modelling thermal system.

(g) Determine the stability of the system represented by the characteristic equation $S^4 + 8S^3 + 18S^2 + 16S + 5 = 0$ using Routh–Hurwitz criteria.

(h) Explain ac tachogenerator.

(i) Draw the ladder diagram for AND operation.

(j) Explain the general block diagram of a microcontroller.

PART B

$(4 \times 10 = 40)$

II. (a) Draw and explain linear variable differential transformer. (5)

(b) Design a pneumatic circuit to sequence two cylinders A and B such that when the start button is pressed, the piston of cylinder A extends and then, when it is fully extended, the piston of cylinder b extends. (5)

OR

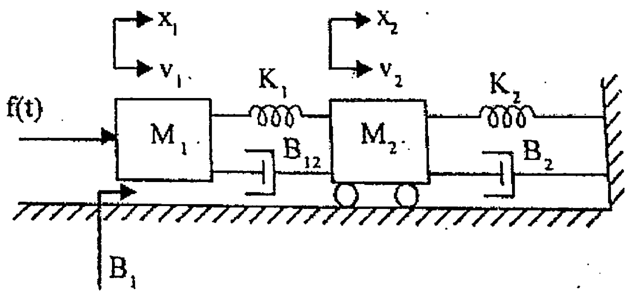
III. (a) Describe the working of an Incremental encoder. (5)

(b) Explain piezoelectric sensors. (5)

(P.T.O.)

IV. (a) Obtain the transfer function of the armature controlled d.c motor. (5)

(b) Draw the force-voltage electrical analogous circuit for the system shown in figure. (5)



OR

V. (a) Obtain the response of a unity feedback system whose open loop transfer function is $G(s) = \dfrac{4}{S(S+5)}$ and when the input is unit step. (6)

(b) What is PID controller and what are its effect on system performance? (4)

VI. Sketch the bode plot for the following transfer function and determine phase margin and gain margin. (10)

$$G(s) = \frac{75(1+0.2s)}{s\left(s^2+16s+100\right)}$$

OR

VII. A unity feedback control system has an open loop transfer function $G(s) = \dfrac{k}{S\left(S^2+4S+13\right)}$, sketch the root locus. (10)

VIII. (a) Draw and explain the internal architecture of a microprocessor. (6)

(b) Explain the system for raising and lowering a barrier for a car park. (4)

OR

IX. (a) Explain pick and place robot. (6)

(b) Draw the ladder diagram and explain the latch circuit. (4)

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